

# SAFETY DATA SHEET

Revision Date 24-December-2021

Revision Number 4

## 1. Identification

**Product Name** 1,2,3,4-Tetrahydroquinoline

**Cat No. :** AC138270000; AC138270250; AC138271000; AC138275000;  
AC138270025

**CAS-No** 635-46-1  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

### Details of the supplier of the safety data sheet

#### Company

##### Importer/Distributor

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

##### Manufacturer

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

## 2. Hazard(s) identification

### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |             |
|---------------------------------------------------------|-------------|
| <b>Skin Corrosion/Irritation</b>                        | Category 2  |
| <b>Serious Eye Damage/Eye Irritation</b>                | Category 2  |
| <b>Carcinogenicity</b>                                  | Category 1B |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3  |
| Target Organs - Respiratory system.                     |             |

### Label Elements

#### Signal Word

Danger

#### Hazard Statements

Causes skin irritation

Causes serious eye irritation  
May cause respiratory irritation  
May cause cancer



### Precautionary Statements

#### Prevention

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

#### Response

IF ON SKIN: Wash with plenty of soap and water  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
IF exposed or concerned: Get medical advice/attention  
If skin irritation occurs: Get medical advice/attention  
Take off contaminated clothing

#### Storage

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

#### Disposal

Dispose of contents/container to an approved waste disposal plant

## 3. Composition/Information on Ingredients

| Component                      | CAS-No   | Weight % |
|--------------------------------|----------|----------|
| Quinoline, 1,2,3,4-tetrahydro- | 635-46-1 | >95      |
| Quinoline                      | 91-22-5  | >0.1-<1  |

## 4. First-aid measures

#### General Advice

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### Skin Contact

Immediate medical attention is required. Wash off immediately with plenty of water for at least 15 minutes.

#### Inhalation

Remove to fresh air. Immediate medical attention is required. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. If not breathing, give artificial respiration.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

|                                        |                                                                                                                                                                                                                                                                                                 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                           |

## 5. Fire-fighting measures

|                                         |                                                                          |
|-----------------------------------------|--------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Dry chemical, Carbon dioxide (CO <sub>2</sub> ), Alcohol resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                 |
| <b>Flash Point</b>                      | 100 °C / 212 °F                                                          |
| <b>Method -</b>                         | No information available                                                 |
| <b>Autoignition Temperature</b>         | No information available                                                 |
| <b>Explosion Limits</b>                 |                                                                          |
| <b>Upper</b>                            | No data available                                                        |
| <b>Lower</b>                            | No data available                                                        |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                 |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                 |

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. In the event of fire and/or explosion do not breathe fumes.

### Hazardous Combustion Products

Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Thermal decomposition can lead to release of irritating gases and vapors.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 3             | 1                   | 0                  | N/A                     |

## 6. Accidental release measures

|                                  |                                                                                                                                                               |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Use personal protective equipment as required. Ensure adequate ventilation. |
| <b>Environmental Precautions</b> | Should not be released into the environment. See Section 12 for additional Ecological Information.                                                            |

**Methods for Containment and Clean Up** Keep in suitable, closed containers for disposal. Soak up with inert absorbent material.

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. Use only under a chemical fume hood. Wear personal protective equipment/face protection. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong oxidizing agents. Strong acids.                                                                                                     |

## 8. Exposure controls / personal protection

**Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment**
**Eye Protection**  
**Hand Protection**

Goggles  
Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers | -               | Splash protection only |
| Neoprene       | recommendations   |                 |                        |
| Natural rubber |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                         |                              |
|-----------------------------------------|------------------------------|
| <b>Physical State</b>                   | Liquid                       |
| <b>Appearance</b>                       | Light yellow                 |
| <b>Odor</b>                             | Odorless                     |
| <b>Odor Threshold</b>                   | No information available     |
| <b>pH</b>                               | 10-11 100 g/l water          |
| <b>Melting Point/Range</b>              | 15 - 17 °C / 59 - 62.6 °F    |
| <b>Boiling Point/Range</b>              | 249 °C / 480.2 °F @ 760 mmHg |
| <b>Flash Point</b>                      | 100 °C / 212 °F              |
| <b>Evaporation Rate</b>                 | No information available     |
| <b>Flammability (solid,gas)</b>         | Not applicable               |
| <b>Flammability or explosive limits</b> |                              |
| <b>Upper</b>                            | No data available            |
| <b>Lower</b>                            | No data available            |
| <b>Vapor Pressure</b>                   | 0.1 mbar @ 20 °C             |
| <b>Vapor Density</b>                    | 4.6                          |
| <b>Specific Gravity</b>                 | 1.060                        |

|                                               |                                  |
|-----------------------------------------------|----------------------------------|
| <b>Solubility</b>                             | No information available         |
| <b>Partition coefficient; n-octanol/water</b> | No data available                |
| <b>Autoignition Temperature</b>               | No information available         |
| <b>Decomposition Temperature</b>              | No information available         |
| <b>Viscosity</b>                              | 12.5 mPa s at 20 °C              |
| <b>Molecular Formula</b>                      | C <sub>9</sub> H <sub>11</sub> N |
| <b>Molecular Weight</b>                       | 133.19                           |

## 10. Stability and reactivity

|                                         |                                                                                                                                                                       |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                                                                                                            |
| <b>Stability</b>                        | Stable under normal conditions.                                                                                                                                       |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat.                                                                                                                                   |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Strong acids                                                                                                                                 |
| <b>Hazardous Decomposition Products</b> | Nitrogen oxides (NO <sub>x</sub> ), Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Thermal decomposition can lead to release of irritating gases and vapors |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                                                              |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                                                                         |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component | LD50 Oral       | LD50 Dermal      | LC50 Inhalation |
|-----------|-----------------|------------------|-----------------|
| Quinoline | 270 mg/kg (Rat) | 1370 mg/kg (Rat) | Not listed      |

**Toxicologically Synergistic Products** No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | No information available                                                                 |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                         | CAS-No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------------------------|----------|------------|------------|------------|------------|------------|
| Quinoline,<br>1,2,3,4-tetrahydro- | 635-46-1 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Quinoline                         | 91-22-5  | Group 2B   | Not listed | Not listed | X          | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated.

**delayed** Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains. .

| Component | Freshwater Algae                                                                                               | Freshwater Fish                                                                                                                                                     | Microtox                         | Water Flea                                                                                   |
|-----------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------------------------------------------------------------------------------------|
| Quinoline | EC50: = 90 mg/L, 96h static (Desmodesmus subspicatus)<br>EC50: = 84 mg/L, 72h static (Desmodesmus subspicatus) | LC50: = 40 mg/L, 96h static (Poecilia reticulata)<br>LC50: = 46 mg/L, 96h static (Pimephales promelas)<br>LC50: = 77.8 mg/L, 96h flow-through (Pimephales promelas) | EC50 34.34 - 130.29 mg/L<br>60 h | EC50: 45.9 - 57.3 mg/L, 48h Static (Daphnia magna)<br>EC50: = 28.5 mg/L, 48h (Daphnia magna) |

**Persistence and Degradability** May persist based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility. Will likely be mobile in the environment due to its water solubility.

| Component | log Pow |
|-----------|---------|
| Quinoline | 2.06    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

**DOT** Not regulated  
**TDG** Not regulated  
**IATA** Not regulated  
**IMDG/IMO** Not regulated

## 15. Regulatory information

### International Inventories

| Component                      | CAS-No   | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|--------------------------------|----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Quinoline, 1,2,3,4-tetrahydro- | 635-46-1 | -   | X    | X    | ACTIVE                                        | 211-237-6 | -      | -   |
| Quinoline                      | 91-22-5  | X   | -    | X    | ACTIVE                                        | 202-051-6 | -      | -   |

| Component                      | CAS-No   | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|--------------------------------|----------|-------|----------|------|------|------|------|-------|-------|
| Quinoline, 1,2,3,4-tetrahydro- | 635-46-1 | X     | KE-33506 | -    | X    | X    | X    | X     | X     |
| Quinoline                      | 91-22-5  | X     | KE-02719 | X    | X    | X    | X    | X     | X     |

### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances  
 KECL - Korean Existing and Evaluated Chemical Substances  
 ENCS - Japanese Existing and New Chemical Substances  
 AICS - Australian Inventory of Chemical Substances  
 PICCS - Philippines Inventory of Chemicals and Chemical Substances

## Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|-----------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Quinoline | Part 2 Substance                                     | Schedule I                                                                 |                                           |

## Other International Regulations

### Authorisation/Restrictions according to EU REACH

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                         | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Quinoline | -                                                                   | Use restricted. See item 72. (see link for restriction details)<br>Use restricted. See item 28. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                      | CAS-No   | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------------------|----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Quinoline, 1,2,3,4-tetrahydro- | 635-46-1 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| Quinoline                      | 91-22-5  | Listed         | Not applicable               | Not applicable            | Not applicable                             |

| Component                      | CAS-No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Quinoline, 1,2,3,4-tetrahydro- | 635-46-1 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Quinoline                      | 91-22-5  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By** Regulatory Affairs  
 Thermo Fisher Scientific  
 Email: EMSDS.RA@thermofisher.com

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**Revision Summary** This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the

date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**